MILITARY SPECIFICATION

CARTRIDGE, 81MM, HE, M374A3
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of high explosive mortar cartridge designated as Cartridge, 81MM, HE, M374A3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-Ammunition and Special Weapons, General Specification for
MIL-C-00401	-Composition B, Military Specification for

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-453	-Inspection, Radiographic
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

PRODUCT AND PACKING DRAWINGS

ARMY

9241291	-Cartridge, 81MM, HE, M374A3 W/Fuze, PD, M524A6
9230175	-Container, Ammo, Fiber, M252A4, For 81MM Mortar Ammunition in Jungle Wrap
9278234	-Box, Wirebound, Packing, Ammo for 81MM Mortar
9287601	-Packing & Marking for Box

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INSPECTION EQUIPMENT DRAWINGS

7257965	-Wear Limit Check
6518071	-Profile and Alignment
6518072	-Alignment Check

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Cartridge.-The cartridge shall comply with all requirements specified on drawing (dwg) 9241291.

3.3 High Explosive Scrap.-High explosive scrap may be used provided that it complies with the requirements for high explosive as specified on (dwg) 8881025 with the exception that it does not have to meet the material form requirement as specified in MIL-C-00401. The high explosive scrap, to be used, shall be clean (see 6.4).

3.4 Cavitation in high explosive charge (see 4.3.1).

3.4.1 General.-Cavities having a maximum (max) projected dimension of 3/32 inch or less shall be disregarded provided that they are separated by 1/2 inch or more of solid explosive.

3.4.2 Maximum projected areas and dimensions of cavities.-The aggregate of the projected area of all cavities shall not exceed 1 square inch and no individual cavity shall exceed 1/2 square inch. Cavities located in the area of the charge which extends from the base of the fuze well to 2.5 inches below the base of the fuze well shall not exceed 3/4 inch in any direction and their aggregate projected area shall not exceed 5/8 square inch.

3.4.3 Piping cavity.-A piping cavity on the longitudinal axis of the projectile shall be permitted provided that its projected length does not exceed 1 1/2 inches and its width does not exceed 1/4 inch. The projected area of such a cavity shall not be included when determining cavity area in the charge.

3.4.4 Surface cavitation.-Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not exceed in aggregate more than 1/4 of the circumference. The max. dimension of any visible cavity in the faced (bottom)

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surface of the fuze well shall not exceed 1/4 inch in any direction and the total area of all cavities shall not exceed 80 percent of the faced surface area. No individual point of explosive shall extend above the flat surface on which the liner would rest. (Cavities having a max. dimension of 1/16 inch or less shall be disregarded in this computation.)

3.4.5 Cracked charges and annular rings.-Not more than four transverse cracks shall be permitted in any charge. The cracks shall be no wider than 1/32 inch. Longitudinal cracks shall be disregarded. The presence of annular rings with a projected width of 1/16 inch max. and extended from the shell wall 1/4 inch max. into the charge shall not be cause for rejection.

3.4.6 Base separation.-The base separation from the inside of the rear of the shell to the base of the explosive shall be not more than 1/32 inch.

3.5 Proving ground test.-The cartridges shall comply with the following requirements when tested in accordance with 4.2.3.4 and 4.3.2.

3.5.1 Misfires.-The cartridges shall not fail to fire from the weapon.

3.5.2 Duds.-The cartridges shall function on impact.

3.5.3 Short rounds.-No cartridge shall have a horizontal range less than 80 percent of the average horizontal range (calculated without the short rounds).

3.5.4 Premature burst.-There shall be no premature burst in the gun bore or in flight.

3.5.5 Metal parts separation.-There shall be no metal parts separation in the gun bore or in flight.

3.5.6 Range standard deviation.-The horizontal range standard deviation shall be no greater than twenty (20) meters at charge zero and no greater than fifty (50) meters at charge four.

3.6 Workmanship.-Parts and assemblies shall be free of burrs, chips, dirt, grease, rust, corrosion and other foreign material.

3.7 Foreign matter in projectile cavity.-There shall be no foreign matter in the projectile cavity prior to high explosive pouring. Solid explosives shall not be present. Pools of paint which have settled, shall be dry and solidified.

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3.8 High explosive on threads.-Threads shall be free of high explosive, after pouring.

3.9 High explosive on projectile body.-Exterior surface of the projectile body shall be free of high explosive after pouring.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisons of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with an identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Quality Conformance Inspection

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4.2.1 Lot formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specification or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Projectile body metal parts assemblies of one lot interfix number, from one manufacturer.
b. Composition B of one lot interfix number from one manufacturer.
c. Fin assemblies, of one lot interfix number, from one manufacturer.
d. Ignition cartridges, of one lot interfix number, from one manufacturer.
e. Propelling charges, made from not more than one lot of propellant, from one manufacturer.
f. Obturating rings, of one lot interfix number, from one manufacturer.
g. Fuzes, of one lot interfix, from one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. An AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Body loading assembly, prior to loading (see dwg. 8881025 covering a detail of dwg. 9241291).

Categories	Defects	Method of Inspection
Critical:		
	1. Foreign object in charge cavity (see 3.7)	Visual
Major:	None defined.	
Minor:	None defined.	

4.2.2.2 Body loading assembly, after loading, but prior to assembly of liner (see dwg. 8881025 covering a detail of dwg. 9241291).

Categories	Defects	Method of Inspection
Critical:		
	1. High explosive on projectile body (see 3.9)	Visual
Major:		
	101. Depth of fuze well, max	Gage CD
	102. Diameter of fuze well, max	Gage CD
Minor:		
	201. Surface cavitation excessive (see 3.4.4)	Visual
	202. High explosive on threads (see 3.8)	Visual
	203. Evidence of poor workmanship (see 3.6)	Visual

4.2.2.3 Body loading assembly, after assembly of liner (see dwg. 8881025 covering a detail of dwg. 9241291).

Categories	Defects	Method Of Inspection
Critical:		
	1. Liner missing	Visual
	2. High explosive extends beyond top of liner.	Visual
Major:		
	101. Depth to bottom of liner, min	Gage CD
Minor:		
	201. Evidence of poor workmanship (see 3.6)	Visual

4.2.2.4 Cartridge (see dwg. 9241291).

Categories	Defects	Method Of Inspection
Critical:		
	1. Ignition cartridge head not fully seated	Visual

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2. Fuze safety wire missing or insecure.....Visual-Manual
3. Propelling charges missing or damaged (cracked, delaminated).....Visual
4. Obturating ring missing.....Visual
5. Fin assembly damaged or distorted.....Visual

Major: 100 percent inspection
101. Obturating ring not welded.....Visual
102. Obturating ring not flush or below bourrelet diameter.....Visual
103. Outside diameter, max.....Gage CD
104. Cartridge assembly fails to freely enter profile and alignment gage.....Gage
Dwg. 6518071
Dwg. 6518072
Dwg. 7257965

Major:
106. Obturating ring damaged.....Visual
107. Fin assembly fails to meet minimum assembly torque.....Test
108. Fuze fails to meet minimum assembly torque.Test
109. Ignition cartridge fails to meet minimum assembly torque.....Test
110. Fin assembly not fully seated.....Visual
111. Fuze not fully seated.....Visual

Minor:
201. Paint damage area in excess of 1/4 square inch.....Visual
202. Pettman cement does not cover a minimum of one half the number of threads for a full 360 degrees (applicable to the fuze, fin and ignition cartridge-prior to assembly)..Visual
203. Marking misleading or unidentifiable.....Visual
204. Evidence of poor workmanship (see 3.6).....Visual

4.2.2.5 Unsealed fiber container (see dwg. 9230175).

Categories Defects Method Of Inspection

Critical: None defined

Major: None defined

Minor:
201. Packing stop missing.....Visual
202. Spacer, cushion or filler disc assembly missing.....Visual
203. Propelling charge supports bind on inside diameter of container preventing easy removal of round.....Manual

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4.2.2.6 Sealed fiber container (see dwg. 9230175).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Container damaged	Visual
102.	Sealing strip incomplete or badly wrinkled.	Visual
103.	Metal end loose or distorted	Visual/Manual
Minor:		
201.	Marking misleading or unidentifiable	Visual
202.	Contents loose	Manual

4.2.2.7 Fiber container in jungle wrap (see dwg. 9230175).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Wax coating incomplete	Visual
102.	Inner wrapper ends improperly folded, exposing fiber container	Visual
Minor:		
201.	Length of sealing strip tab	Gage CD
202.	Marking missing or unidentifiable	Visual

4.2.2.8 Sealed wirebound box (see dwg 9278234).

Categories	Defects	Method of Inspection
Minor:		
201.	Boards split or separated exposing containers	Visual
202.	Contents loose	Manual
203.	Marking misleading or undentifiable	Visual
204.	Strap insecure	Manual/Visual
205.	Wire broken	Visual
206.	Wireloops improperly fastened	Visual

4.2.3 Testing

4.2.3.1 Check test for possible deterioration of Cartridge, Ignition, XM299 (see Specification MIL-C-48156).-If the total elapsed time between original acceptance of the Ignition Cartridge lot and the assembly of that lot into complete round cartridges exceeds two years, or if the Ignition Cartridges have been

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subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-C-48156 immediately before the Ignition Cartridge lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges.

4.2.3.2 Cavitation (see 3.3 - except 3.3.4) Major defect.- The sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-1235 using a CSP-2 plan having an AQL of 0.65 percent. Appropriate i and f values shall be as specified by the procuring activity. Any cartridge having cavitation in excess of that permitted by the applicable requirement for any segment shall be classed defective. The test shall be performed as specified in 4.3.1.

4.2.3.3 Strength of heat weld (see dwg. 8881025) Major defect.-One obturating ring sample, minimum, after every 350 cartridges, shall be assembled to a fixture, introduced to the production line, heat sealed and tested as specified in 4.3.3. Failure of the sample to meet the requirement shall be cause for those 350 preceding cartridges represented by the sample to be rebanded and retested.

4.2.3.4 Proving ground test.-The cartridges shall be observed for the following characteristics. Testing shall be in accordance with 4.3.2:

a. Misfires, major defect (see 3.5.1)
b. Duds, major defect (see 3.5.2)
c. Short rounds, critical defect (see 3.5.3)
d. Premature burst, critical defect (see 3.5.4)
e. Metal parts separation, critical defect (see 3.5.5)
f. Range standard deviation, major defect (see 3.5.6)

4.2.3.4.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, eighty random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. two or more misfires occur
b. four or more duds occur
c. one or more short rounds occur
d. one or more premature bursts occur
e. one or more metal parts separations occur
f. horizontal range standard deviation at charge 0 or charge 4, multiplied by the factor 0.79 exceeds the applicable requirement.

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4.2.3.4.2 After three consecutive lots have complied with the criteria of 4.2.3.3.1, forty eight random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. two or more misfires occur
b. four or more duds occur
c. one or more short rounds occur
d. one or more premature bursts occur
e. one or more metal parts separations occur
f. horizontal range standard deviation at charge 0 or charge 4, multiplied by the factor 0.75 exceeds the applicable requirement.

4.2.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2.

4.3 Test Methods and Procedures

4.3.1 Determination of cavities.-The sample cartridges (HE filled) shall be subjected to radiographic examination. The radiographic beam shall form a 90° angle with the longitudinal axis of the projectile. Examinations of the radiograph shall be made to determine compliance with applicable requirements. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. In addition, the sensitivity of the radiographic technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25, as determined in accordance with 4.3.1.2. In the event an original radiograph leaves doubt as to acceptability, the cartridge shall be re-radiographed at an angle 90° from that of the original exposure and in the same plane.

4.3.1.1 Sensitivity check of radiographic technique.-The sensitivity of radiographic technique shall be determined by the means of a penetrameter fabricated as specified herein. Location of the penetrameter hole shall be 6 3/4 inches measured from base shoulder of projectile body towards fuze nose threads. The projectile, with attached penetrameter, shall be positioned between the radiographic source and the radiographic film with the penetrameter facing the radiographic source. When radiographing a group of projectiles on a rotating ring, the penetrameter, taped to a projectile, may be used to represent the group when checking sensitivity. The sensitivity of radiographic technique shall be considered satisfactory when the image of the penetrameter, including the holes, is sharply defined on the radiograph.

The penetrameters shall be made of steel having an approximate specific gravity of 7.8. Penetrameter thickness shall be .020 plus or minus (±) .010 inches. Each pene-

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trameter shall have a circular hole with a diameter twice that of the penetrameter thickness. For further details on the construction of the penetrameters, see Standard MIL-STD-453. Frequency of sensitivity check shall be no less than one in each fifty (50) radiographs.

4.3.1.2 Check of photographic density of film.-The photographic density of the film shall be determined on the film in the region of the maximum-equivalent thickness to be radiographed. Film density shall be checked each hour of operation, or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. To judge densities, it is recommended that standards be used. Comparison of the film with standards may be made visually; however, the use of a densitometer is recommended.

4.3.2 Proving ground test.-The test shall be performed at a Government Proving Ground in accordance with the applicable USATECOM Acceptance Test Procedure.

The cartridges shall be fired from a standard 81mm mortar at an elevation of forty-five degrees. One half of the sample cartridges shall be fired at charge one and the other half at charge nine. All fuzes shall be set for superquick functioning. The individual velocities, pressures and ranges shall be recorded on the firing record. Observation shall be made for misfires and duds, and shall be recorded. In the event of a misfire, the ignition cartridge shall be removed and forwarded to Picatinny Arsenal. Misfired ignition cartridges shall not be disassembled by the proving ground. Should a premature burst or metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The velocity and range standard deviations and arithmetic averages shall be calculated and recorded. Range probable error shall be calculated and recorded. Temperature of the day, weather and the time of firing shall also be recorded.

Hawk radar shall be used to observe for metal parts separation, unobserved short rounds and unobserved duds. In these events, an analysis should be made of the radar data and should be noted on the firing record. This shall include a comparison of times of flight of the unobserved rounds with that of the normal rounds. Prints of the radar traces of those rounds, and prints of the preceeding and succeeding rounds, shall be attached to the firing record. Data from ballistically acceptable lots may be discarded. Smear camera shall be used to photograph all samples. The camera shall be positioned to pick up each round just outside the second velocity coil. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all the other rounds in that particular charge zone, shall be attached to the firing record. Film from acceptable lots may be discarded.

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4.3.2.1 Test validity.-If for any reason the proving ground or Picatinny Arsenal considers that the test conditions have detrimentally affected the test results, the test results shall be declared invalid and a new test shall be performed on additional sample cartridges.

4.3.3 Strength of heat weld.-The sample shall be tested for integrity by applying the required load at the heat weld radially and in a direction away from the axis (center) of the ring.

5. PREPARATION FOR DELIVERY

5.1 Level of packaging and packing.-Packaging and packing shall conform to Level A, as established in DOD Instructions 4100.14 "Unform Preservation - Packaging, Packing and Marking of Items of Supply".

5.2 Packaging, packing and marking.-Packaging, packing and marking shall be in accordance with dwg. 9230175 and dwg. 9287601.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection equipment designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and Contractor Designs. SIE designs are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the Contractor. Equipment to be designed by the Contractor is designated "CD". The Contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the Contractor may submit alternate or modified Contractor designs of SIE in accordance with 6.2.2 and 6.2.3, should he elect to do so.

6.2.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

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b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the Contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified, and may include commercial equipment which the Contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs, if not otherwise covered thereby (See 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The Contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each Contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion data for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 Submission of test data.-One copy of proving ground ballistic acceptance test data (i.e., firing record) will be forwarded to the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-A, SARPA-AD-E-A4
Dover, New Jersey 07801

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6.4 Clean scrap.-Clean scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation. Scrap from floor sweepings, ventilations, and suction apparatus (other than that of drilling operations) is not defined as being clean scrap and shall not be used.

6.5 Loading of high explosive.-Solid explosive shall not be added to the melted explosive or the the projectile body during pouring or cooling.

6.6 Proving ground test summary.

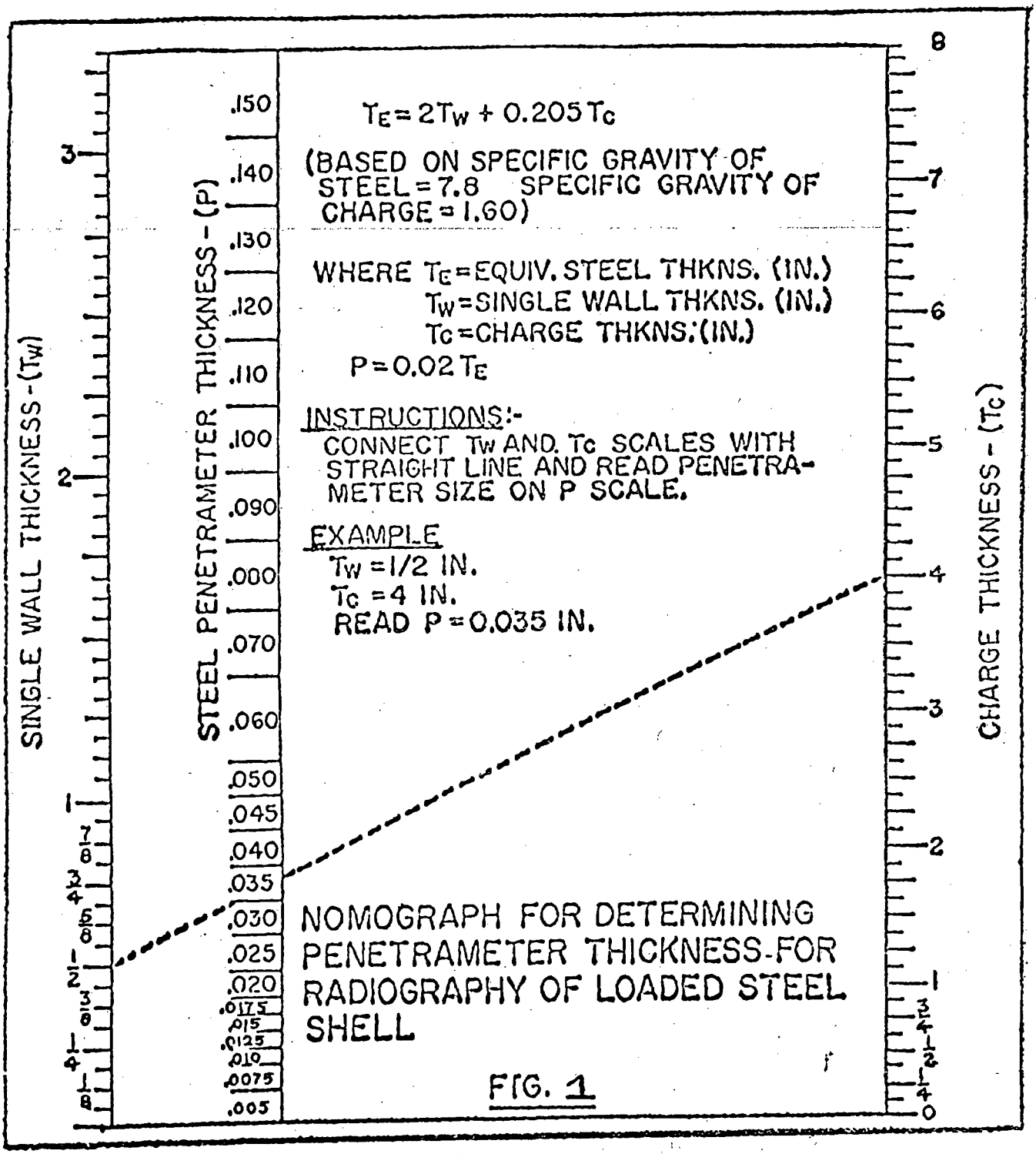
TEST	SAMPLE SIZE	REQUIREMENTS
Ballistic	(see 4.2.3.4) Eighty samples until three consecutive lots have complied with the acceptance criteria. Forty-eight samples thereafter.	a. Misfires (see 3.5.1) b. Duds (see 3.5.2) c. Short rounds (see 3.5.3) d. Premature Burst (see 3.5.4) e. Metal Parts Separation (see 3.5.5) f. Range Standard Deviation (see 3.5.6)

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 1315-A-024

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$T_E = 2T_W + 0.205\,T_C$

(BASED ON SPECIFIC GRAVITY OF STEEL = 7.8 SPECIFIC GRAVITY OF CHARGE = 1.60)

WHERE T_E = EQUIV. STEEL THKNS. (IN.)
T_W = SINGLE WALL THKNS. (IN.)
T_C = CHARGE THKNS. (IN.)

$P = 0.02\,T_E$

INSTRUCTIONS:-
CONNECT T_W AND T_C SCALES WITH STRAIGHT LINE AND READ PENETRAMETER SIZE ON P SCALE.

EXAMPLE
T_W = 1/2 IN.
T_C = 4 IN.
READ P = 0.035 IN.

NOMOGRAPH FOR DETERMINING PENETRAMETER THICKNESS-FOR RADIOGRAPHY OF LOADED STEEL SHELL

FIG. 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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